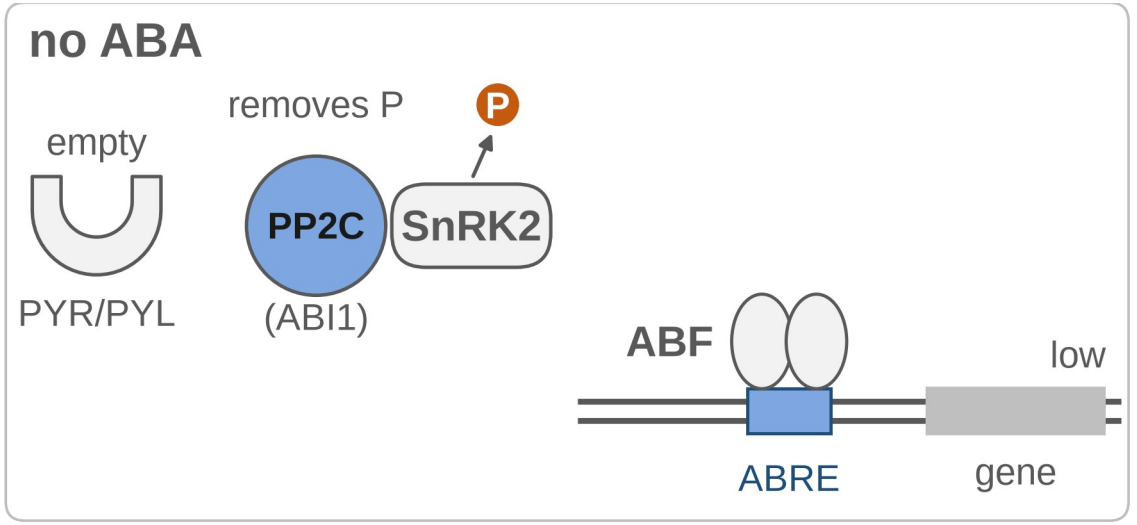
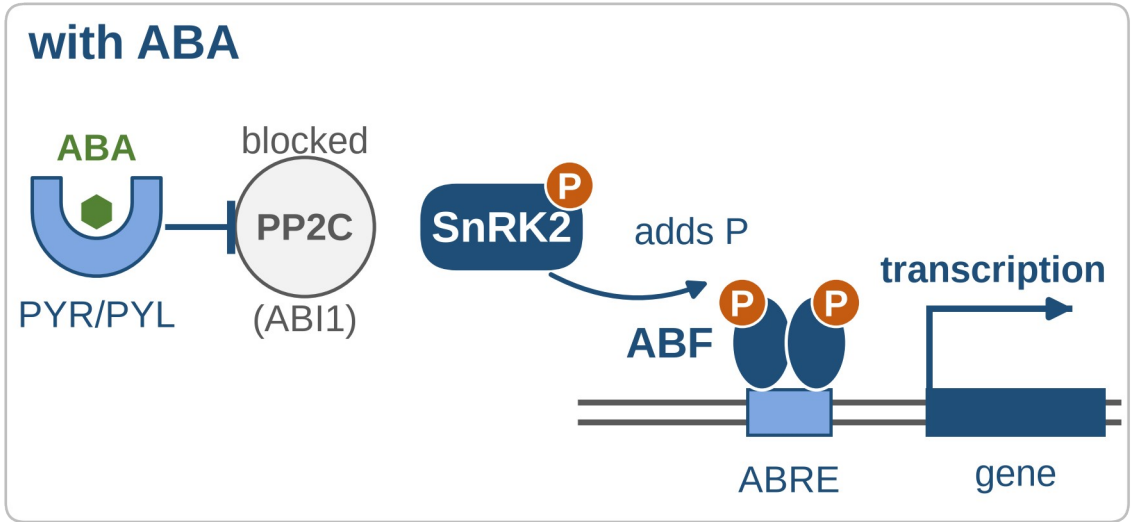
The biology and the experiment

ABA switches on stress genes

- Drought, salt, cold → plants make **abscisic acid (ABA)**
- ABA receptor blocks the phosphatase **ABI1** (PP2C)
- Kinases (**SnRK2**) become active
- They phosphorylate **ABF** transcription factors
- ABFs bind the **ABRE** in promoters → genes on

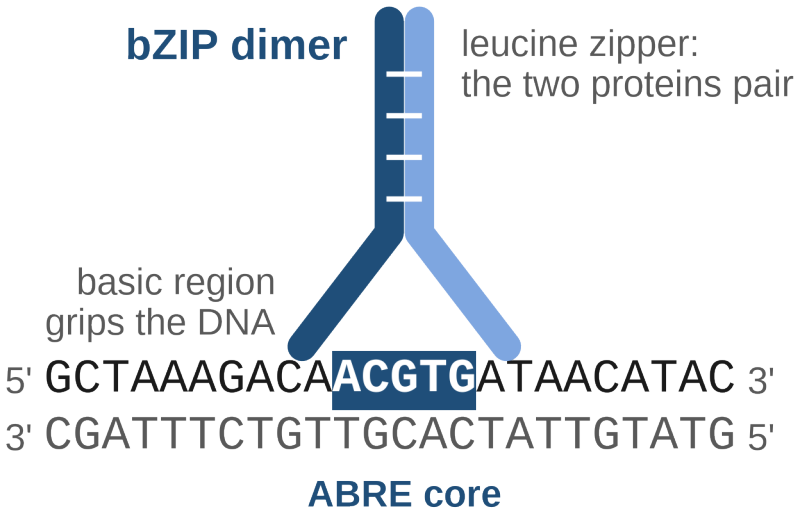




Cutler et al. 2010, Annu Rev Plant Biol 61:651; Fujii et al. 2009, Nature 462:660

ABF3 binds the ABRE as a dimer

- ABFs are **bZIP** proteins: a basic DNA-binding region + a leucine zipper
- Two bZIPs pair (dimer) and grip the DNA
- **ABRE**: a short motif with the core ACGTG
- ABF1–4 have very similar DNA-binding regions



the same core in five promoters (toy)

geneA	ACG**ACGTG**ACGAAACTTGTTGGCCCAGT	+ strand
geneB	GTGAATCGCTTAAGGGT**CACGT**AAGTGT	− strand
geneC	ACCCCATCGGA**ACGTG**ATTTTTATTACA	+ strand
geneD	CTCAGAAACAGAACTCGGGTA**ACGTG**GA	+ strand
geneE	AGTGCG**CACGT**ACTCGCTATGAATCTCT	− strand

CACGT on the + strand = ACGTG read on the − strand

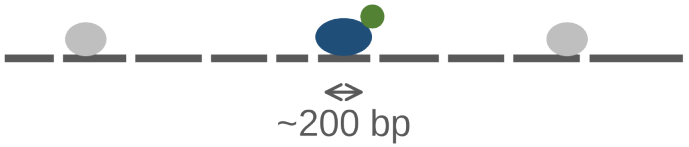
Fujita et al. 2011, J Plant Res 124:509; Yoshida et al. 2010, Plant J 61:672

ChIP-seq: pull down, sequence, map

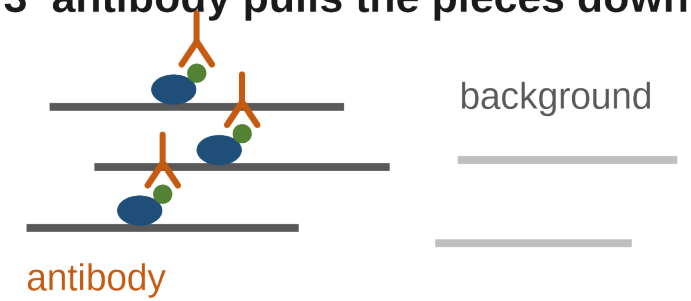
1 tagged protein on its site



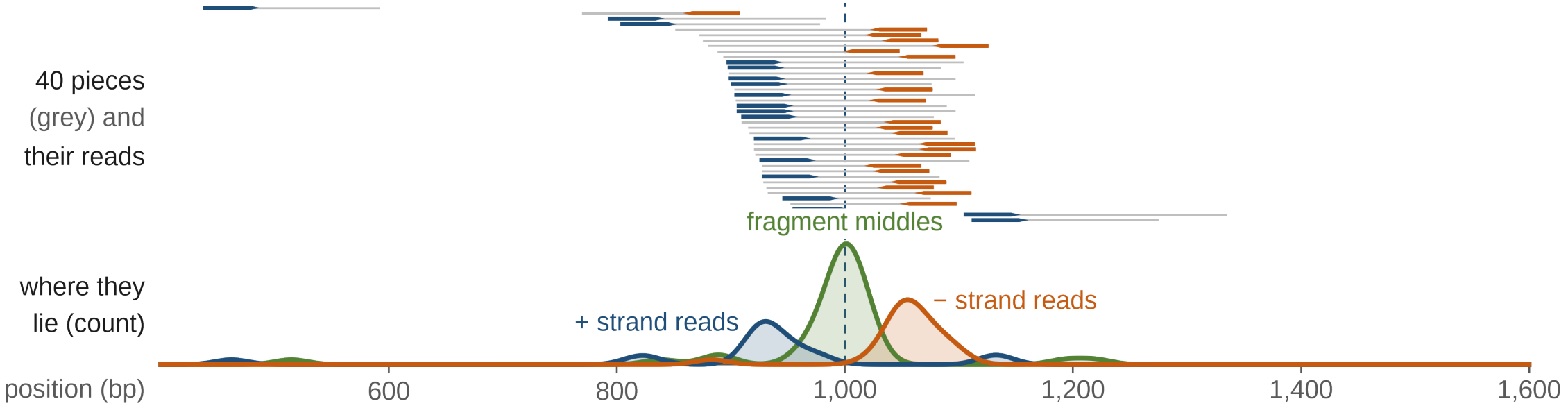
2 cross-link, cut to ~200-bp pieces



3 antibody pulls the pieces down

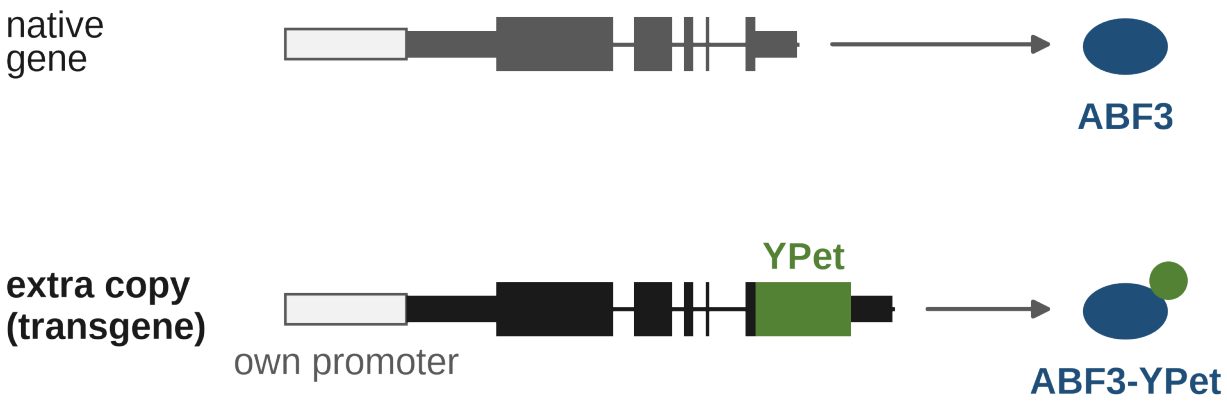


4 sequence 50 bp from one end of each piece, map to the genome



ABF3-YPet: a tag instead of an antibody

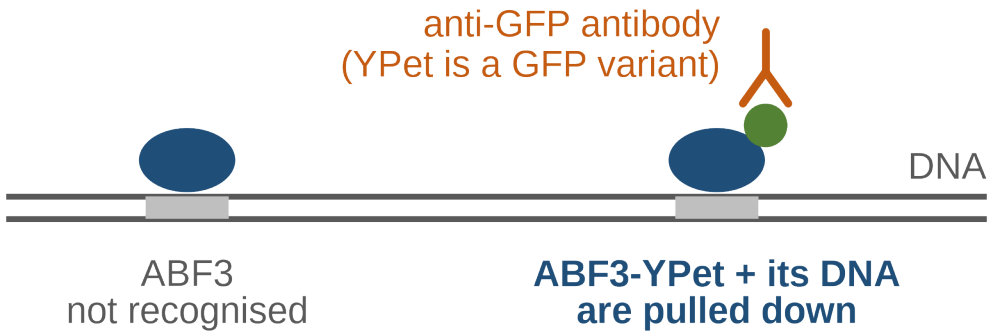
the ABF3-YPet plant has two copies of ABF3



same promoter: both proteins are made
in the same cells at the same time

why a tag? good antibodies against single plant TFs are rare;
with a tag, one antibody works for every TF (21 in the study)

ChIP with an anti-GFP antibody

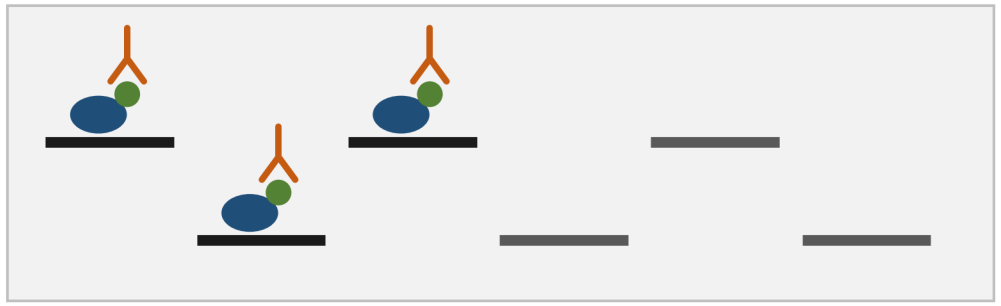


the ChIP finds where the tagged ABF3 sits;
we assume the tag does not change where ABF3 binds

Song et al. 2016, Science 354:aag1550

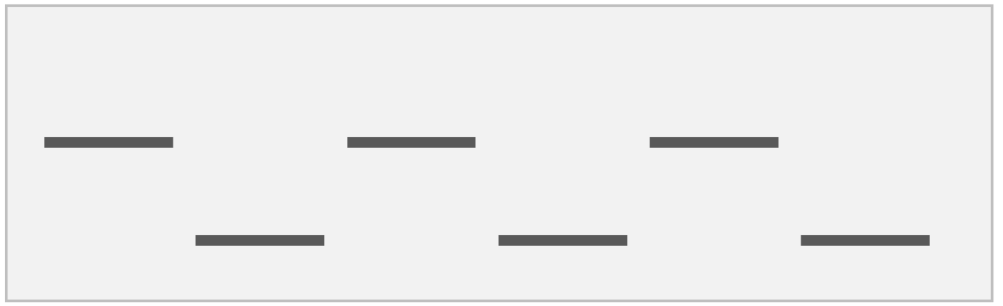
The control: same ChIP, nothing to catch

ABF3-YPet plants + anti-GFP



pulled down: ABF3 sites + background

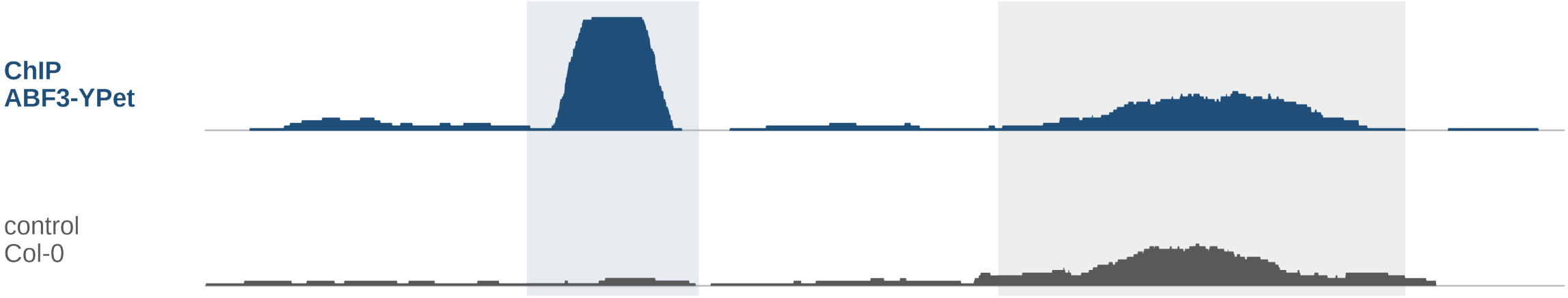
Col-0 (no YPet) + anti-GFP



pulled down: background only

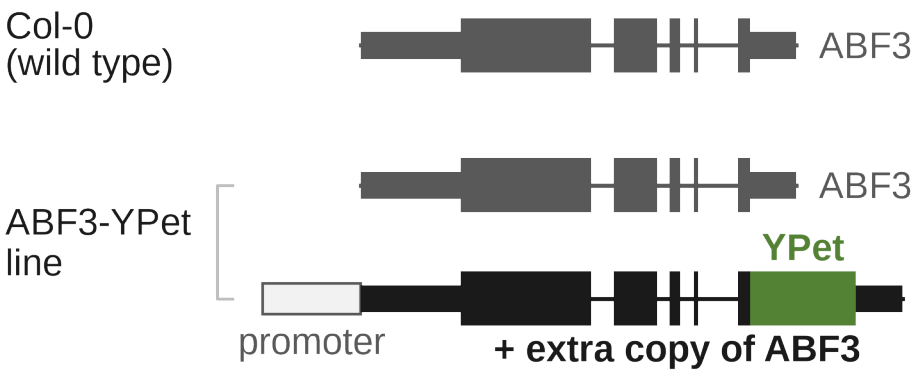
binding site: ChIP >> control

background: high in both - not a binding site

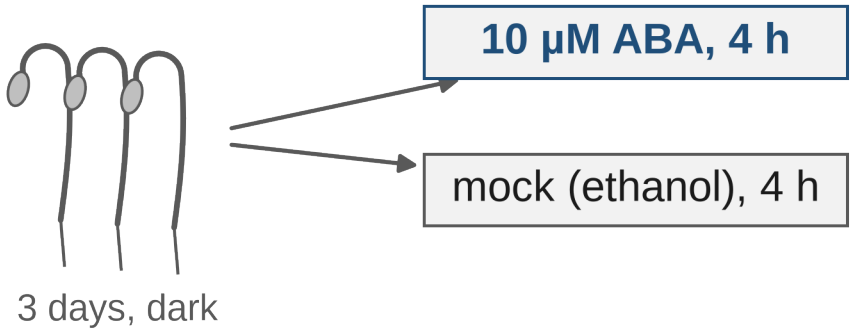


Treatment and samples

plants



treatment



ChIP: anti-GFP antibody (it also binds YPet)

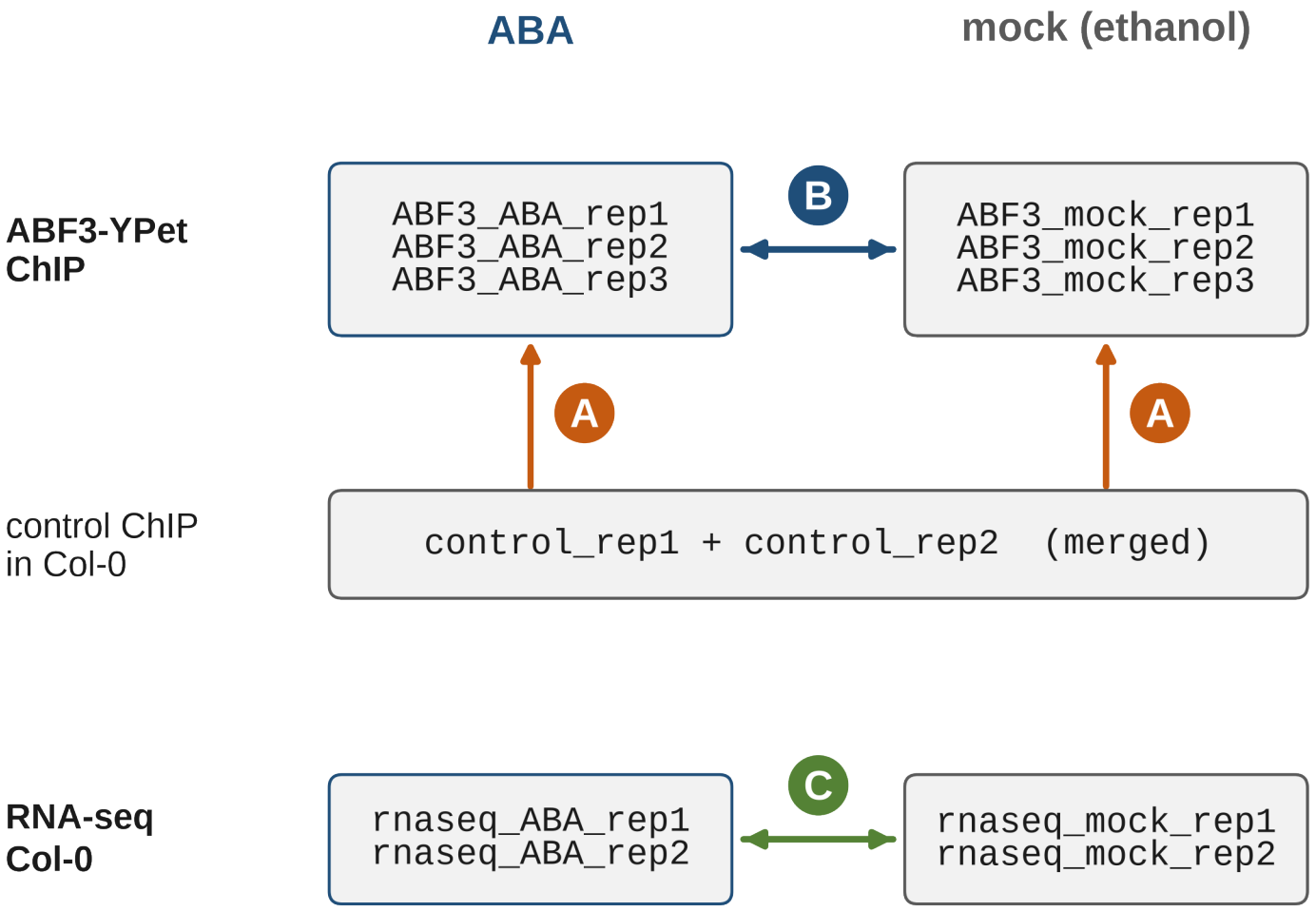
samples (file names)

	ABA 4 h	mock 4 h
ChIP ABF3-YPet line	ABF3_ABA_rep1 ABF3_ABA_rep2 ABF3_ABA_rep3	ABF3_mock_rep1 ABF3_mock_rep2 ABF3_mock_rep3
control: ChIP in Col-0	control_rep1 control_rep2	
RNA-seq Col-0	rnaseq_ABA_rep1 rnaseq_ABA_rep2	rnaseq_mock_rep1 rnaseq_mock_rep2

all reads: 50 bp, single-end
ChIP 2.6–3 M reads, RNA-seq 4 M reads per sample

Song et al. 2016, Science 354:aag1550 (GEO GSE80564, GSE80565)

What is compared with what



A **ChIP vs control**

where does ABF3 bind?

peaks: MACS3

B **ChIP: ABA vs mock**

does the binding change?

DiffBind

C **RNA-seq: ABA vs mock**

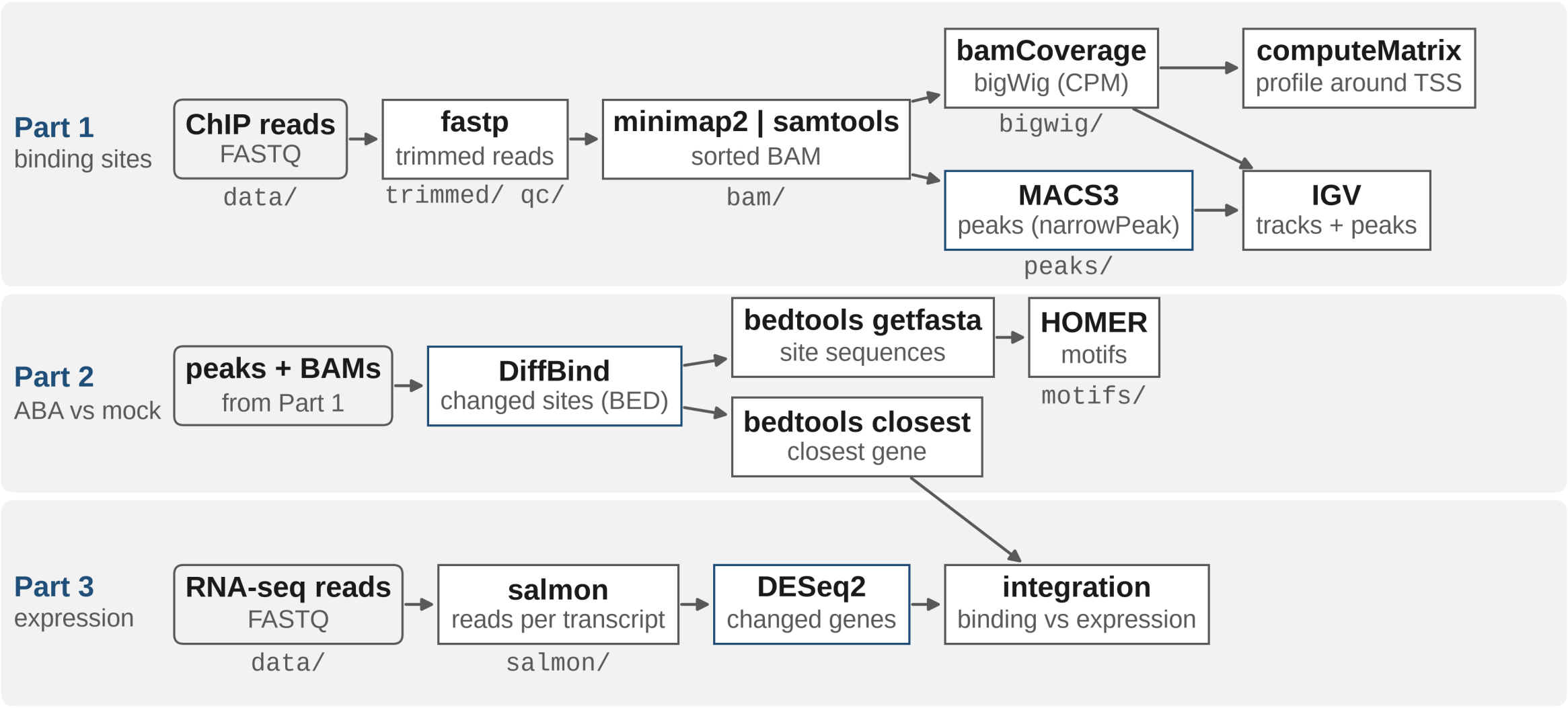
which genes change?

DESeq2

B + C are the genes with gained binding induced?

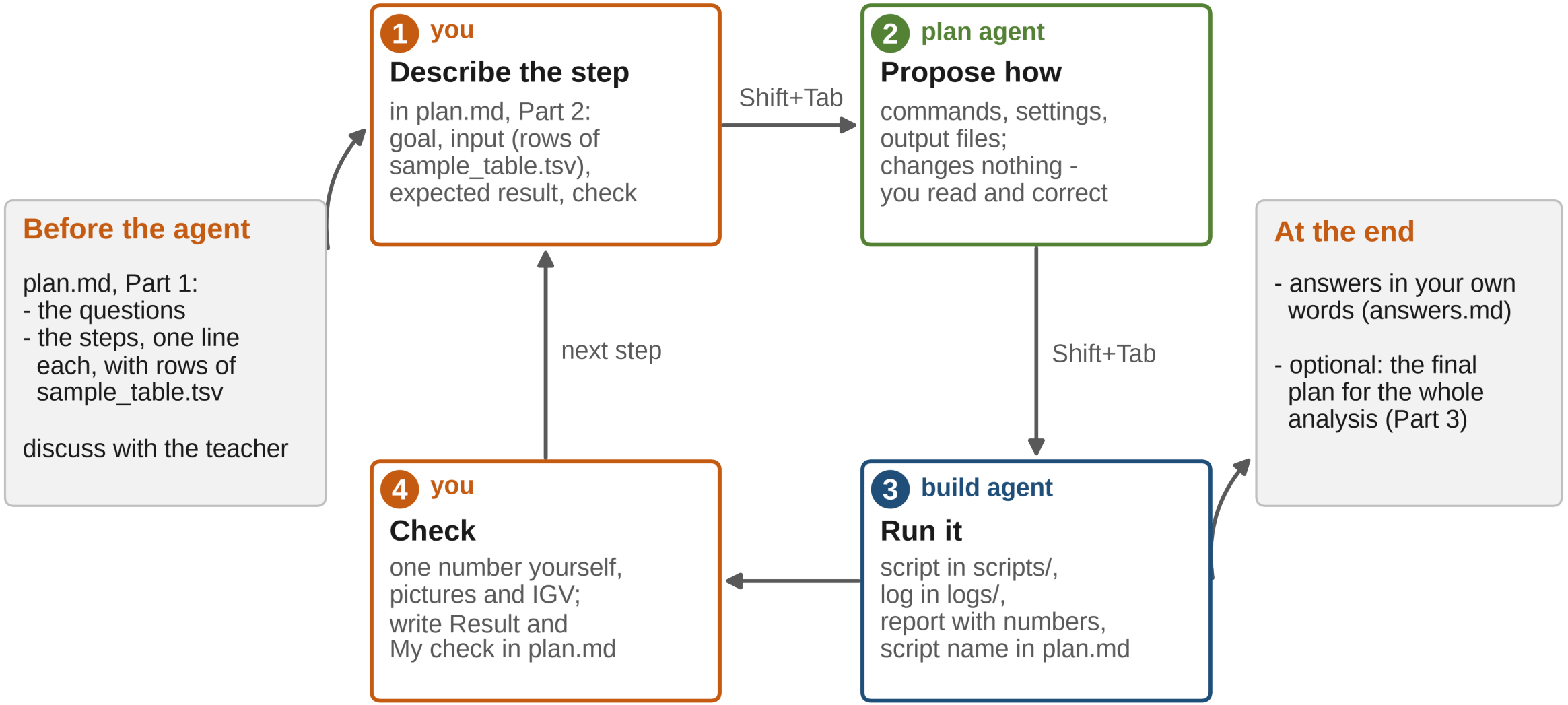
the control is never compared with ABA or mock directly

Three parts, one question



Working with the agent

Plan first, then one step at a time



Two agents, a few keys

- **plan** agent: reads files and proposes - changes nothing → you *think*
- **build** agent: writes scripts and runs commands → it *does*
- `Shift+Tab` switches between them - look which one is active before you send
- `Esc` stops the agent; `/export` saves the conversation (hand-in)
- Refer to files, do not paste: *"Read plan.md, Step 2. Propose how to do it."*
- plan.md open in VS Code with **File** → **Auto Save**: you see what the agent writes

Describe the step, not the commands

weak

```
## Step 2 - Mapping
Goal: do the mapping
```

the agent must guess: which samples? which output? what to report?

good

```
## Step 2 - Mapping
Goal: map the trimmed reads to TAIR10;
  % mapped and reads on Mt and Pt
  for every sample
Input: trimmed reads of the rows with
  assay = ChIP-seq or ChIP-seq control;
  merge the two control rows into one
Expected result: most reads map
How I check it: samtools flagstat of
  one BAM file myself
```

Goal: what the agent reports back. Expected, check: for you.

sample_table.tsv

file	assay	treatment
ABF3_ABA_rep1.fastq.gz	ChIP-seq	ABA
ABF3_ABA_rep2.fastq.gz	ChIP-seq	ABA
ABF3_ABA_rep3.fastq.gz	ChIP-seq	ABA
ABF3_mock_rep1.fastq.gz	ChIP-seq	mock
ABF3_mock_rep2.fastq.gz	ChIP-seq	mock
ABF3_mock_rep3.fastq.gz	ChIP-seq	mock
control_rep1.fastq.gz	ChIP-seq control	not stated
control_rep2.fastq.gz	ChIP-seq control	not stated
rnaseq_ABA_rep1.fastq.gz	RNA-seq	ABA
rnaseq_ABA_rep2.fastq.gz	RNA-seq	ABA
rnaseq_mock_rep1.fastq.gz	RNA-seq	mock
rnaseq_mock_rep2.fastq.gz	RNA-seq	mock

8 rows selected - the agent finds the files here

Part 1 – From reads to binding sites

fastp: what to read in the report

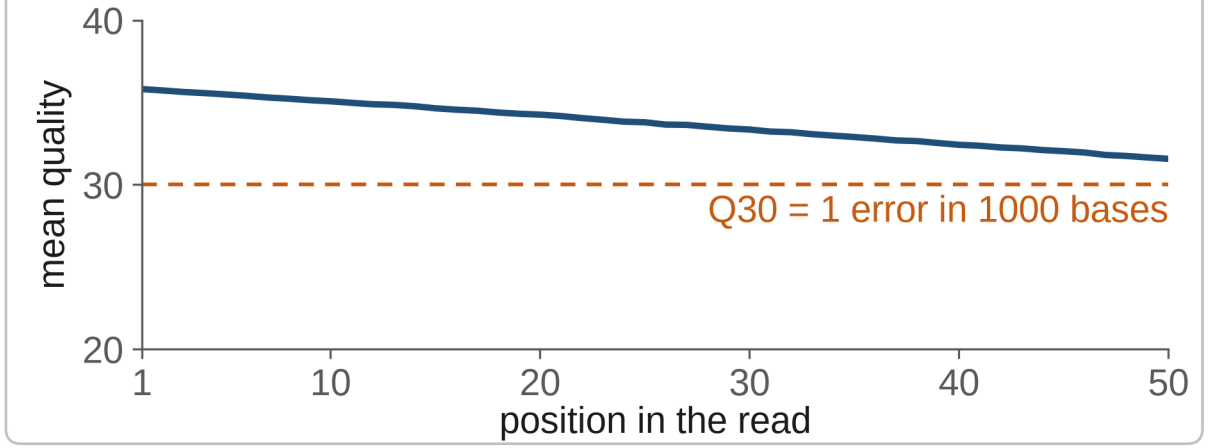
Summary

	Before filtering	After filtering
total reads	20,000	19,989
Q30 bases	90.2 %	90.2 %
mean length	50 bp	50 bp

duplication rate: 8.3 %

reads with low quality: 11 (removed)

Before filtering: read1: quality



duplicate = exactly the same sequence as another read

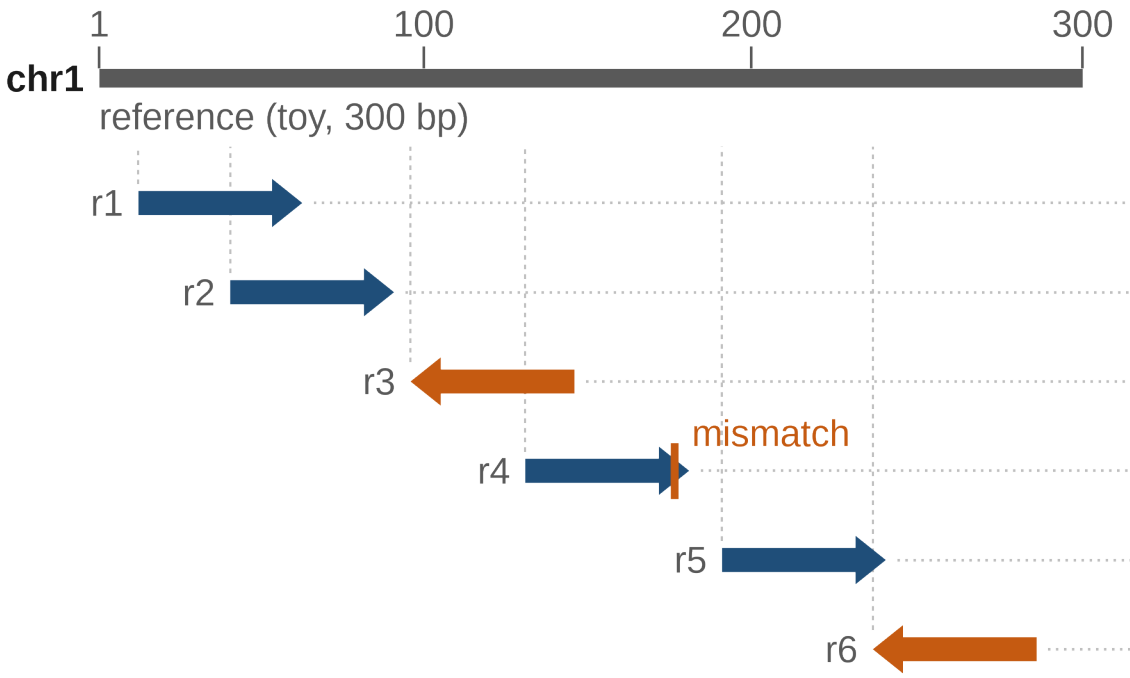
```
read1  TTTCCTCATGCAATTCAAAACCATGTCCGTAATGTAGGCGAAATAGTAAA
read2  CCATTTTACGGAGGATACCAAATTCCTCCTTATTCAGGACCTAACCTGAG
read3  GTAAACCAGGTCTCTCCGCCCCCTTATAAAAGCTGTTGCACCTAGCCAAG
read4  TTCAACGGCAGCTGCAATGGAAATAGGCAATGACGGATATATATTAAAAA
read5  GTGTTTTAAGATACATTGAGGCCCGTTCGTGCTCCTCGCCCTGAAGCATT
read6  GCTTTGTGAAGAGGGACTTCAGCCAATAGACCTGCATACCGGCTCATTCT
read7  CCATTTTACGGAGGATACCAAATTCCTCCTTATTCAGGACCTAACCTGAG  = read2
read8  TCTAATAATATACATTTGCTTCGTTGACTAGCAACCCAGGGCTATAGCTA
read9  TTCCCCCCGCGGCCCACCCAGTATTCCTAACGGAGCATAAATCCCACCCG
read10 GTGTTTTAAGATACATTGAGGCCCGTTCGTGCTCCTCGCCCTGAAGCATT  = read5
```

2 duplicates / 10 reads

= 20 %

fastp on these 10 reads:
duplication rate 20.0 %

Mapping: reads to positions



QNAME	FLAG	RNAME	POS	MAPQ	CIGAR
r1	0	chr1	13	34	50M
r2	0	chr1	41	13	50M
r3	16	chr1	96	37	50M
r4	0	chr1	131	22	50M
r5	0	chr1	191	26	50M
r6	16	chr1	237	20	50M

FLAG 0 = read on the + strand (→)

FLAG 16 = read on the − strand (←)

POS = first reference base covered by the read

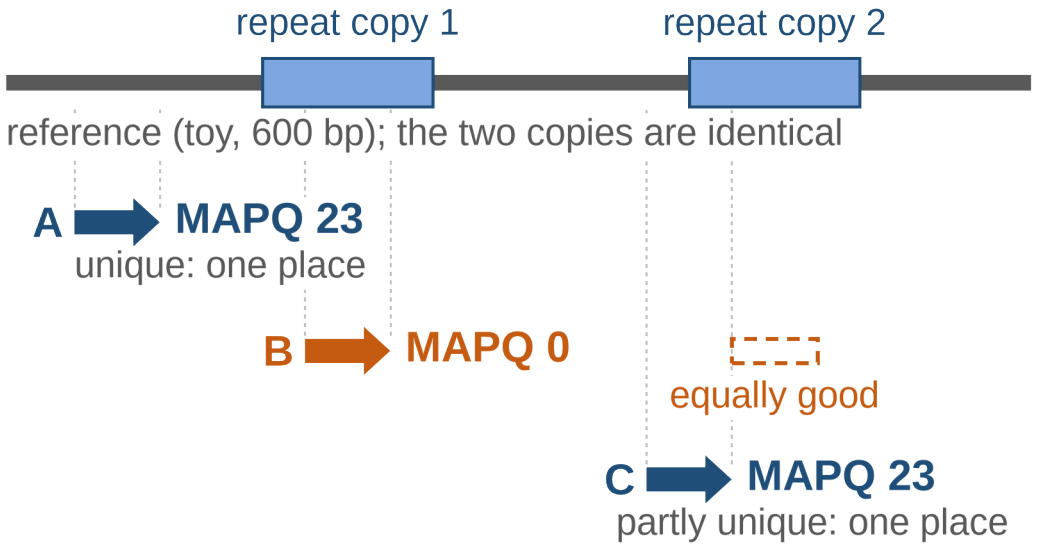
CIGAR 50M = 50 aligned bases

(M = match or mismatch: r4 is 50M too)

MAPQ = how certain the position is

MAPQ: is the position certain?

- A read from a **repeat** fits several places equally
- The mapper picks one at random and reports **MAPQ ≈ 0**
- Unique read → MAPQ > 0 (minimap2 maximum 60; short reads rarely reach it)
- `--minMappingQuality 10` drops the uncertain ones



QNAME	FLAG	RNAME	POS	MAPQ	CIGAR
A	0	chr1	41	**23**	50M
B	0	chr1	176	**0**	50M
C	0	chr1	376	**23**	50M

--minMappingQuality 10 keeps A and C, drops B

samtools idxstats: reads per chromosome

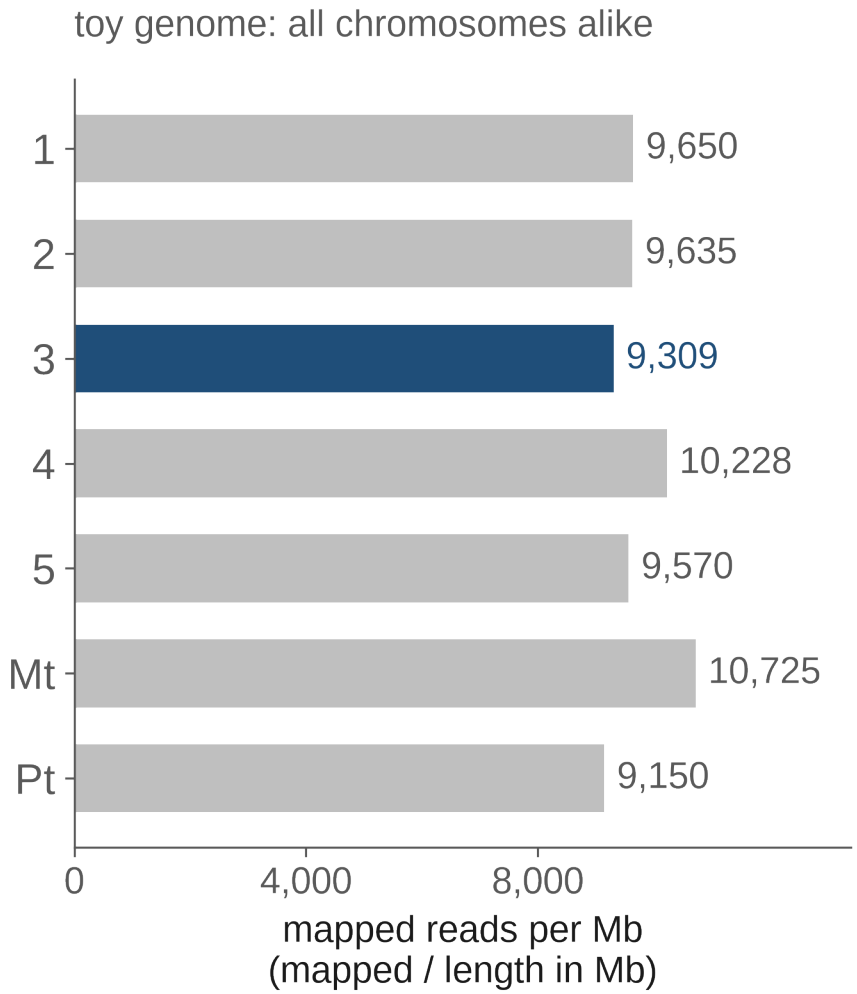
```
$ samtools idxstats sample.bam
```

name	length	mapped	unmapped	
1	300000	2895	0	
2	200000	1927	0	
3	230000	2141	0	
4	180000	1841	0	
5	270000	2584	0	
Mt	40000	429	0	
Pt	20000	183	0	
*	0	0	400	← reads without a position

chromosome 3:
share of the mapped reads

2,141 / 12,000 = 17.8 %

12,000 = sum of the mapped column (without the * line)



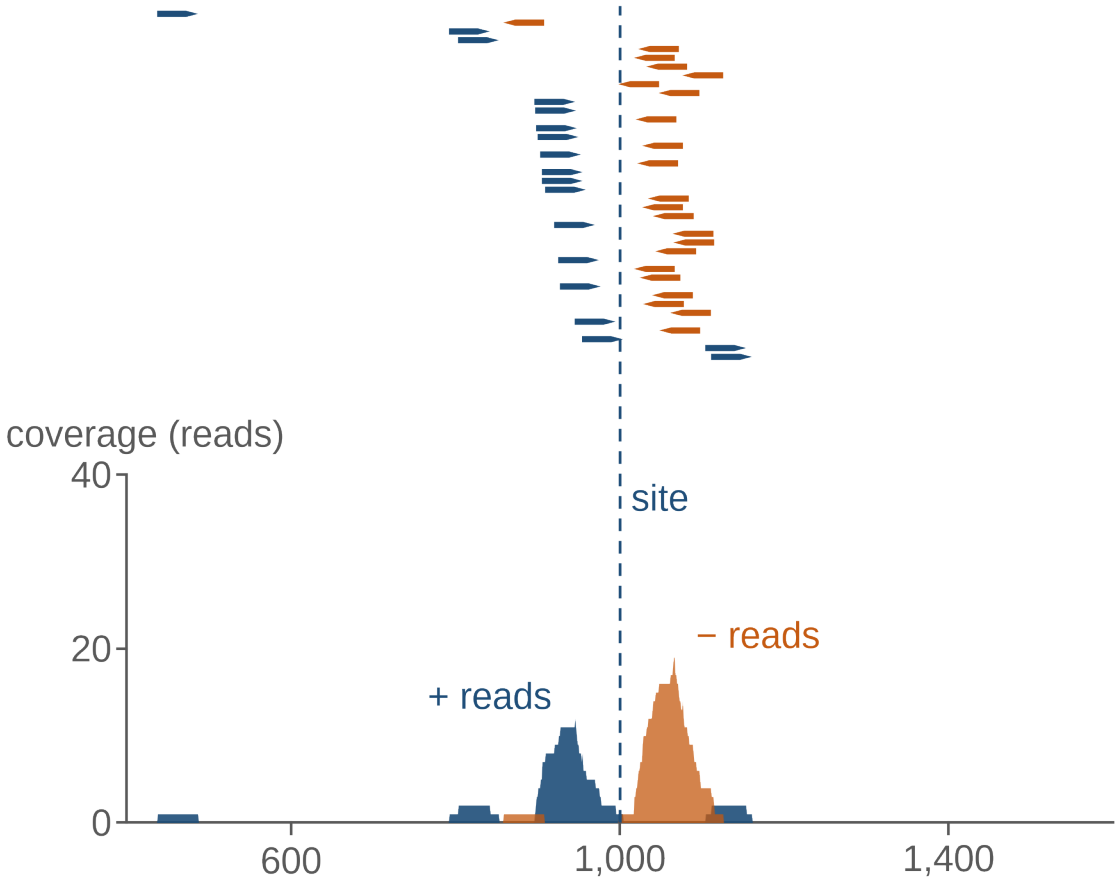

```
$ samtools flagstat sample.bam
12400 + 0 in total (QC-passed reads + QC-failed reads)
12000 + 0 mapped (96.77% : N/A)
```

12,000 / 12,400 = 96.8 % of all reads mapped

Reads are fragment ends

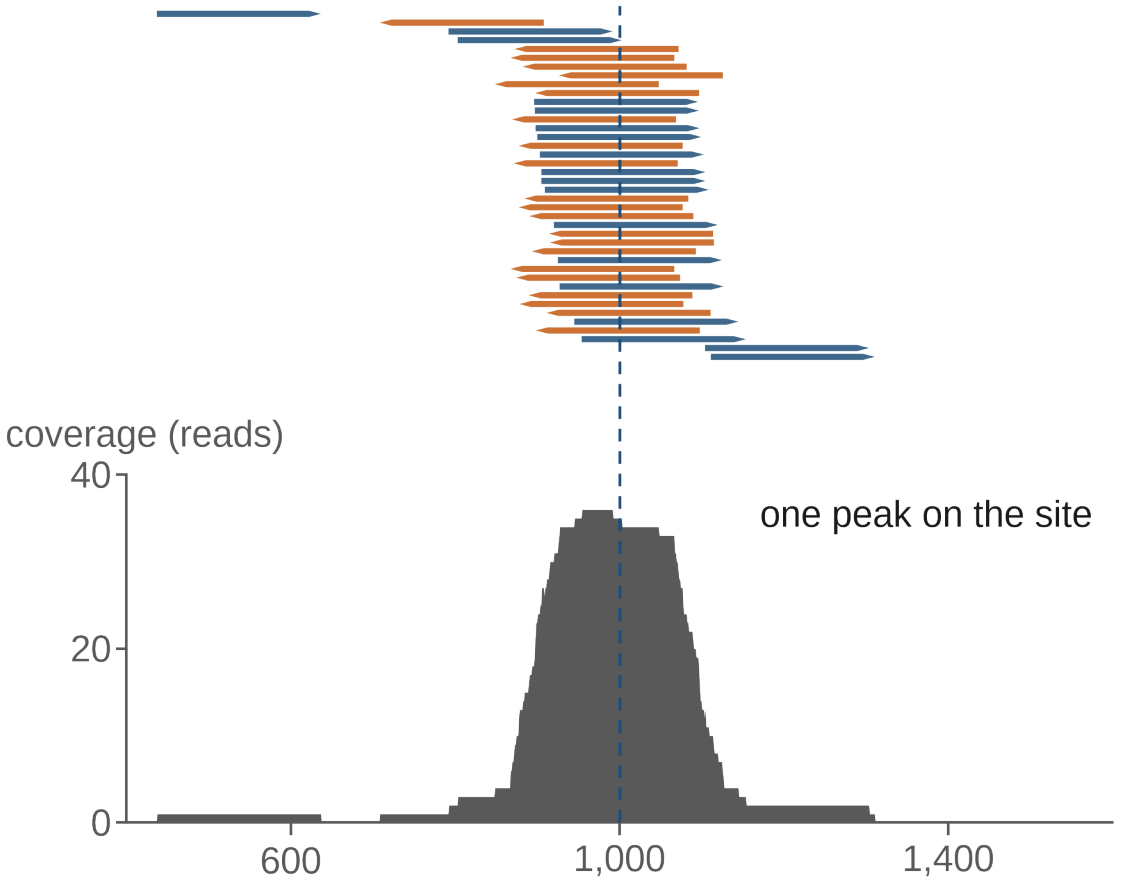
Reads as mapped (50 bp)

two humps, either side of the site



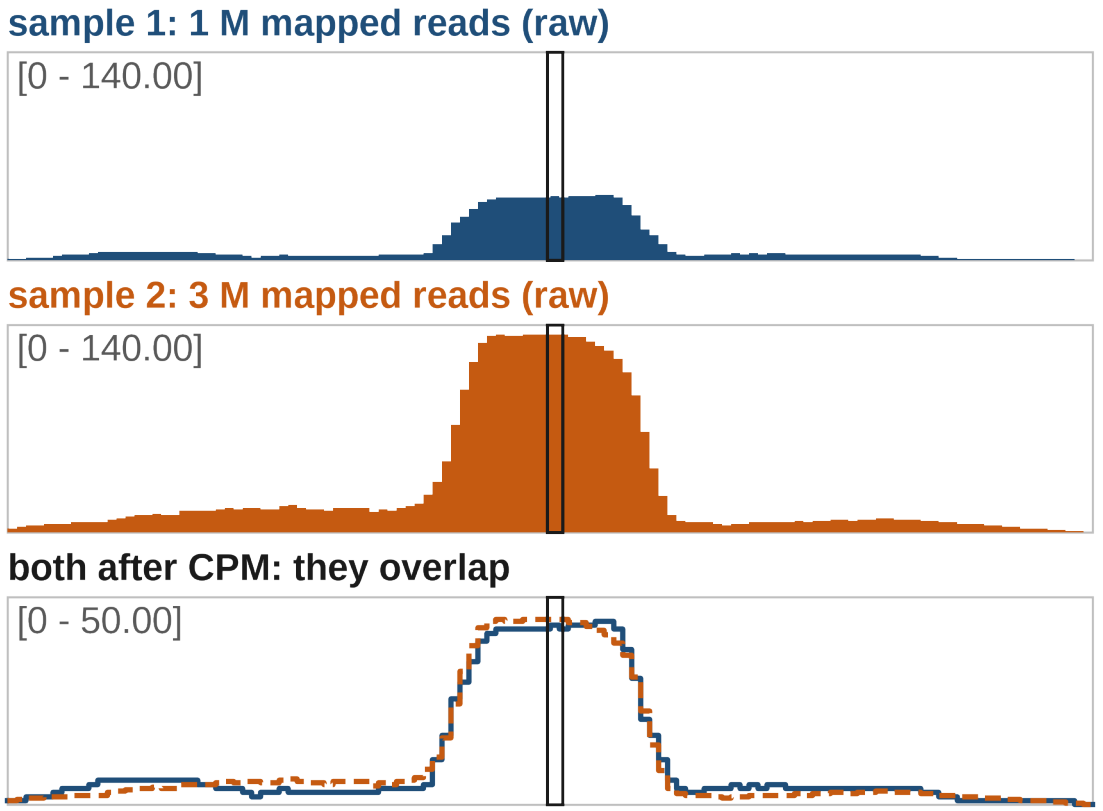
Each read extended to 200 bp

`--extendReads 200`



CPM: comparing samples of different depth

- Sample with 3× more reads → 3× higher coverage everywhere
- **CPM** = reads / million mapped reads
- After CPM, equal binding looks equal
- bigWig = coverage in bins (here 10 bp), small and fast



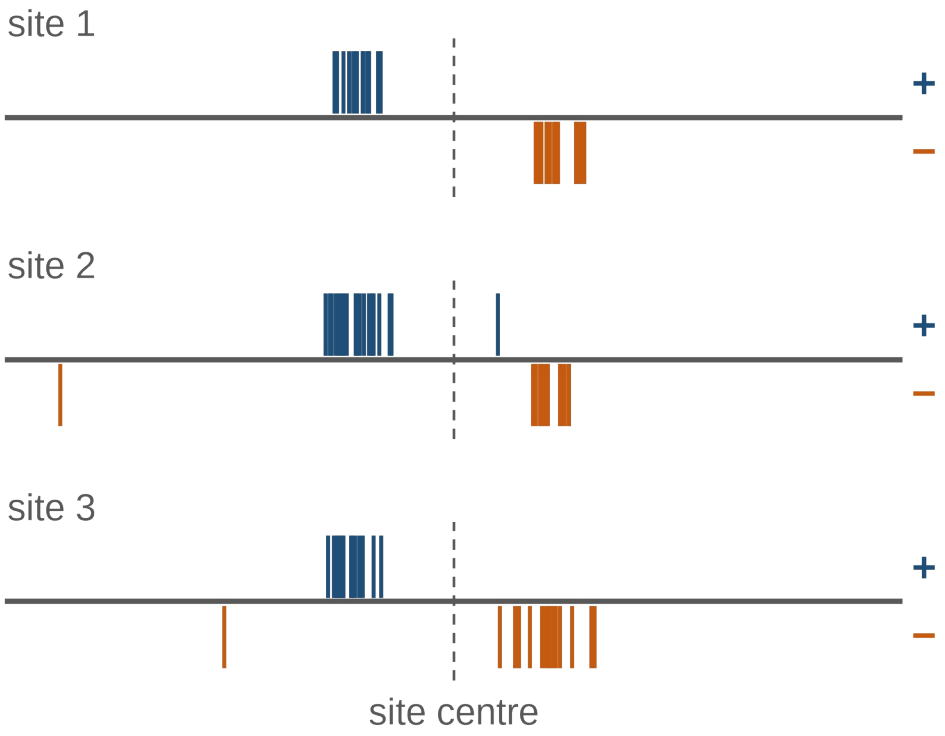
bin 1,000–1,010:

sample 1: 44 reads / 1 M = 44.0 CPM

sample 2: 136 reads / 3 M = 45.3 CPM

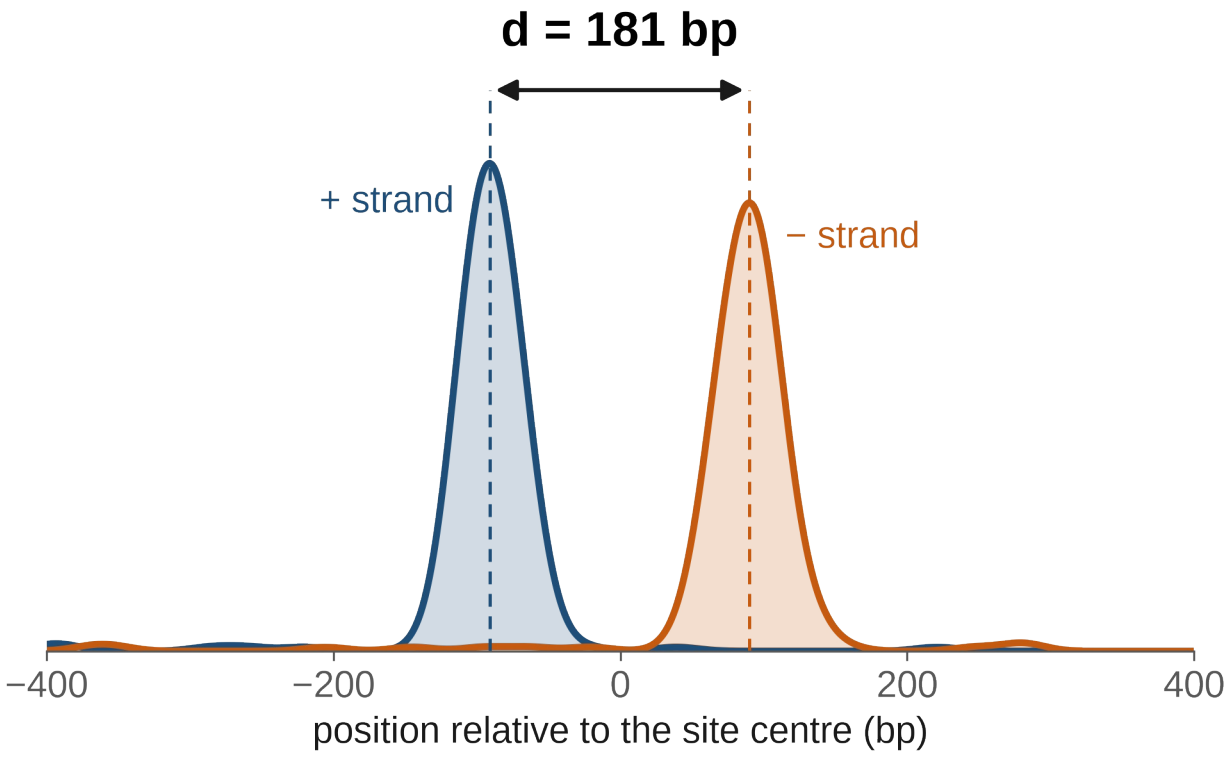
MACS3: estimating the fragment length

Read 5' ends at single strong sites



… and so on for many sites

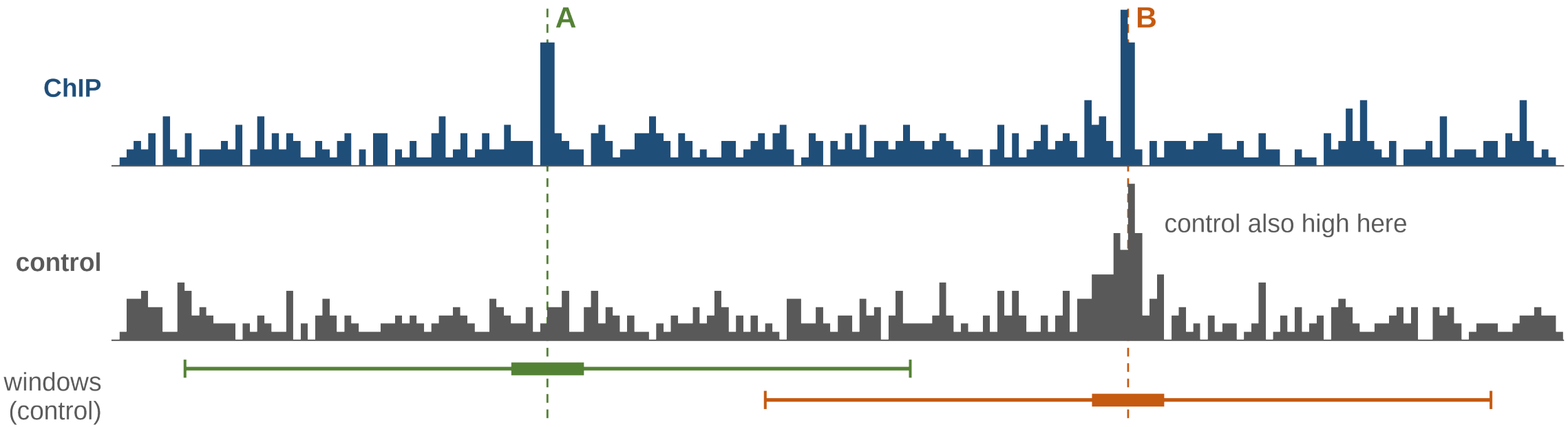
Averaged over 12 sites




```
NAME_peaks.xls:   # d = 181
```

Zhang et al. 2008, Genome Biol 9:R137

MACS3: more reads than expected?



	ChIP reads	λ genome	λ 1 kb	λ 10 kb	λ local	p	q	peak?
A	30	5.0	5.4	5.0	5.4	2e-13	2e-11	yes
B	34	5.0	19.2	6.3	19.2	0.0014	0.071	no

per 200 bp window

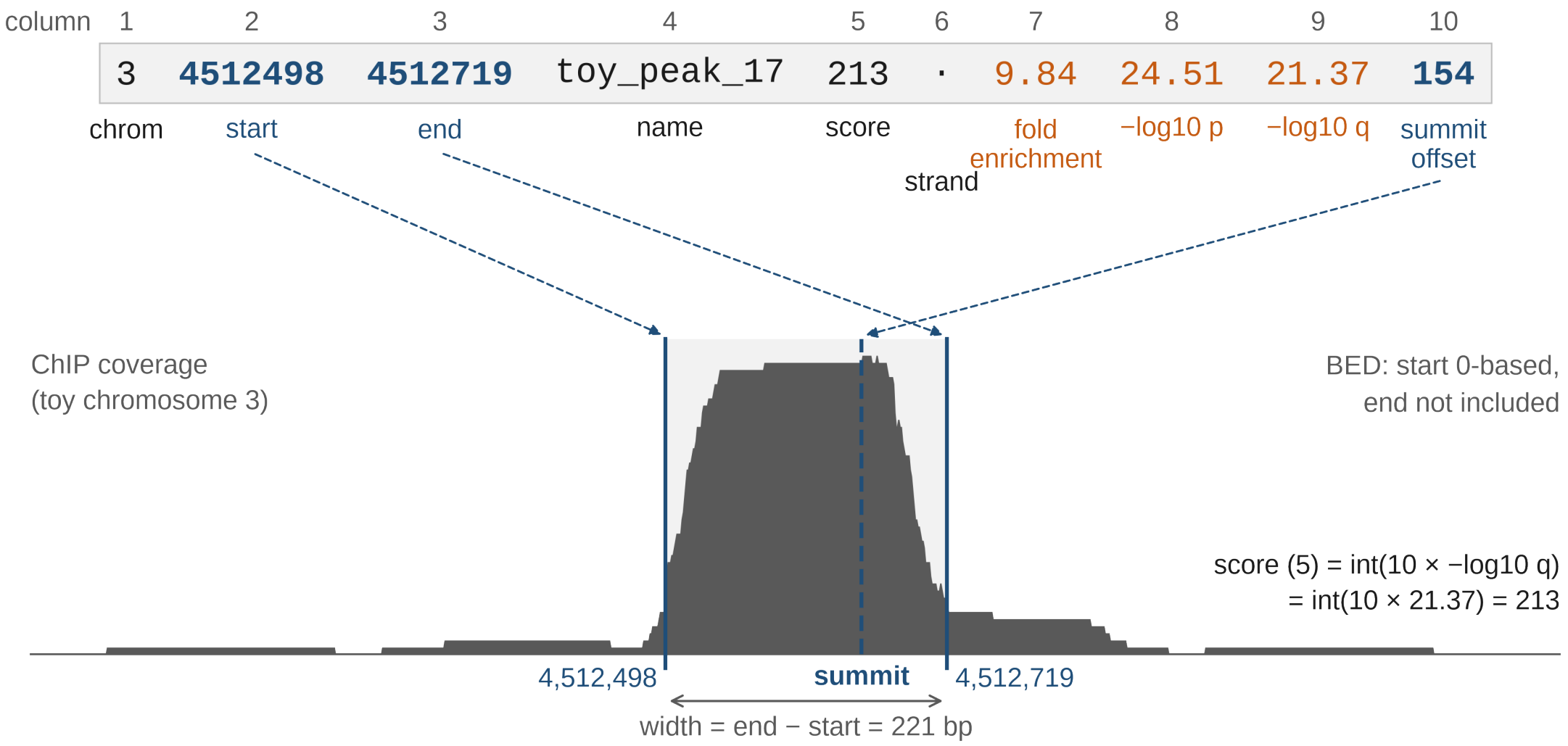
λ local = the largest

peak if q < 0.01

λ genome = 3 M reads × 200 / 1.2e8 (-g) = 5.0 p = Poisson probability of ≥ that many reads q = p corrected for many windows

Zhang et al. 2008, Genome Biol 9:R137; MACS3 callpeak documentation

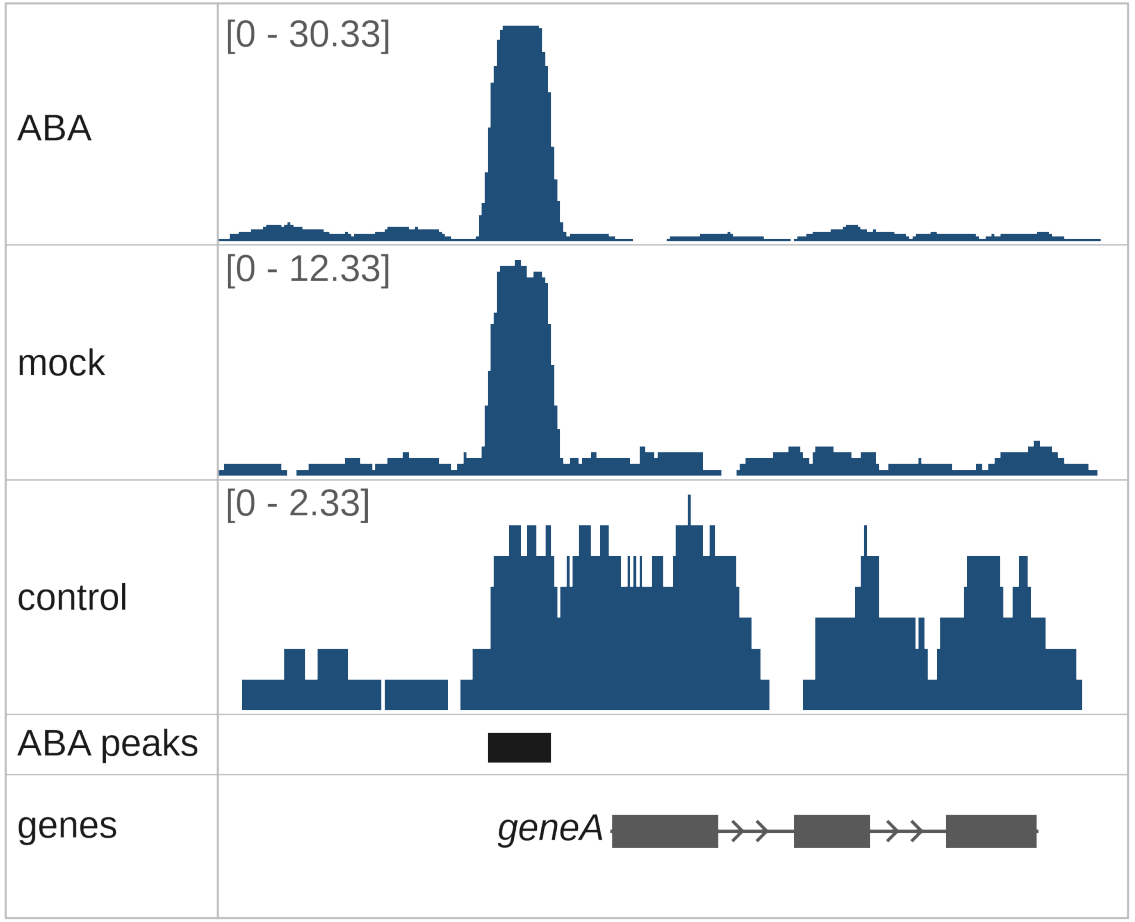
A narrowPeak line



IGV: compare tracks on one scale

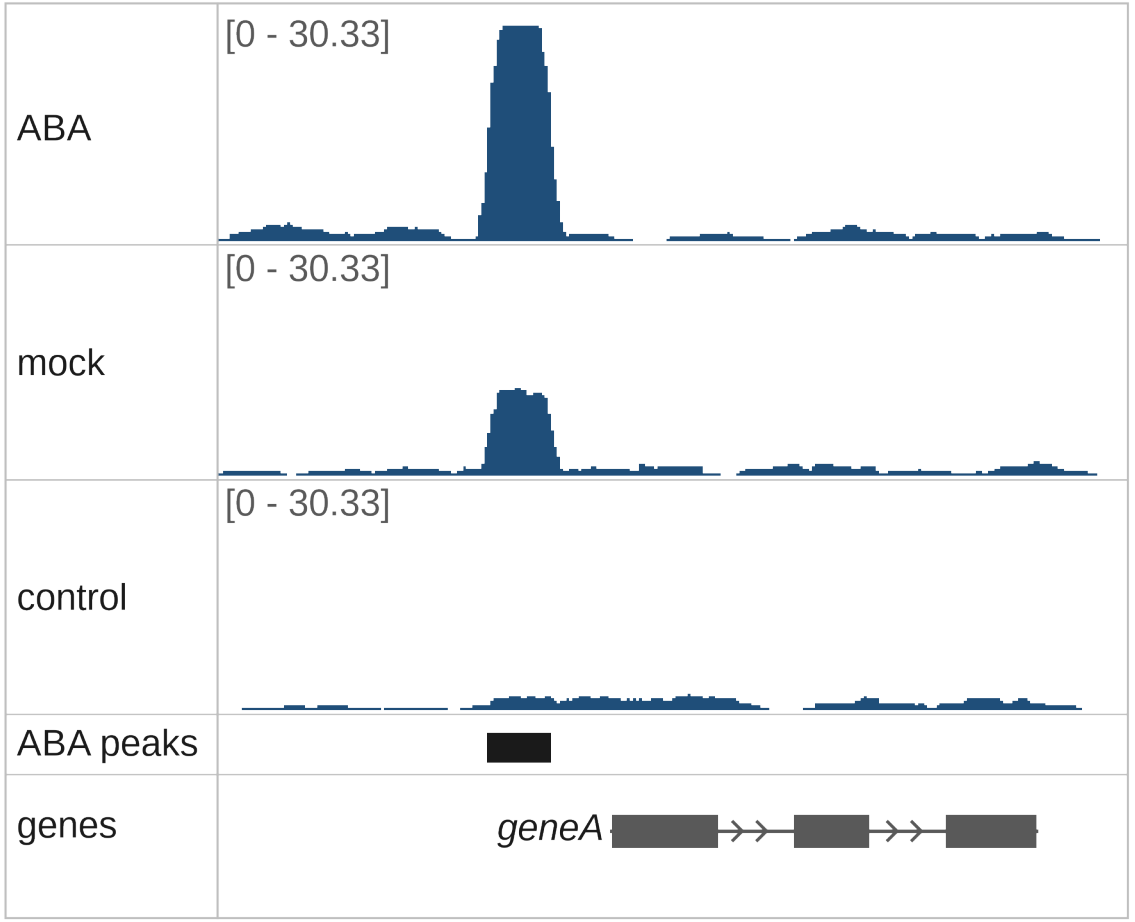
Autoscale (each track alone)

IGV default: every track fills its height

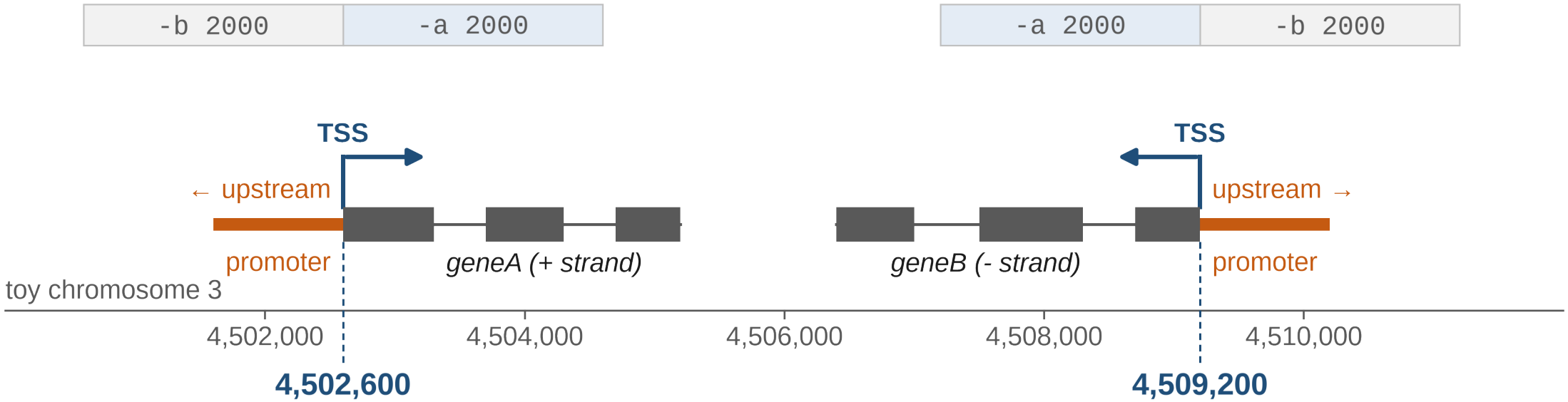


Group Autoscale (one range)

select tracks → right-click → Group Autoscale



Genes have a direction

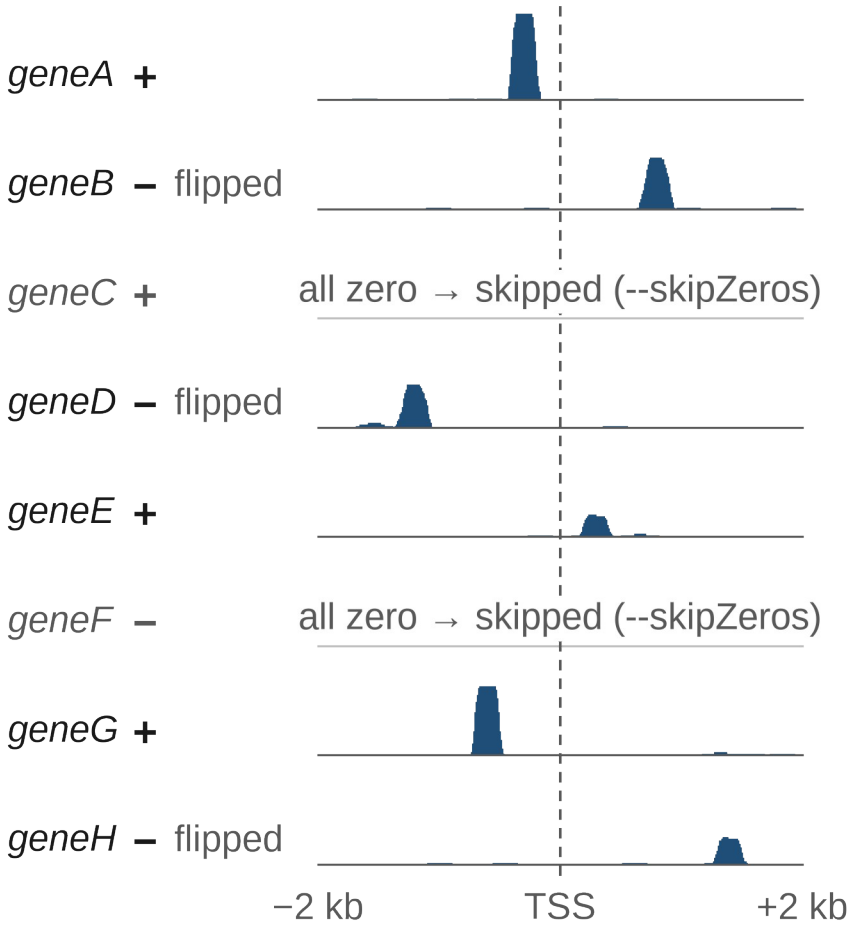


TAIR10_genes.bed-style lines

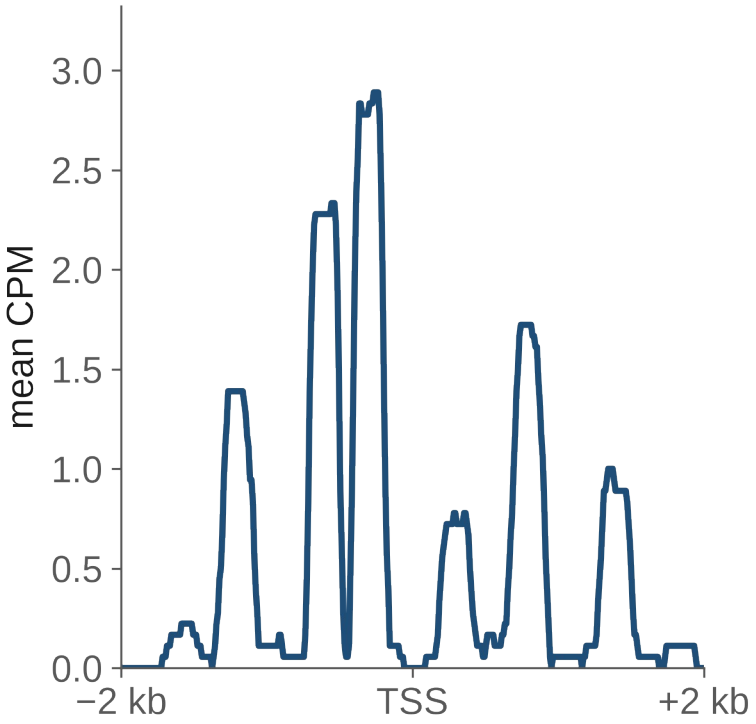
3	**4502600**	4505200	AT3G99010	geneA	**+**	TSS = start
3	4506400	**4509200**	AT3G99020	geneB	**-**	TSS = end

computeMatrix: signal around many genes

1 window around each TSS
(− strand genes flipped)

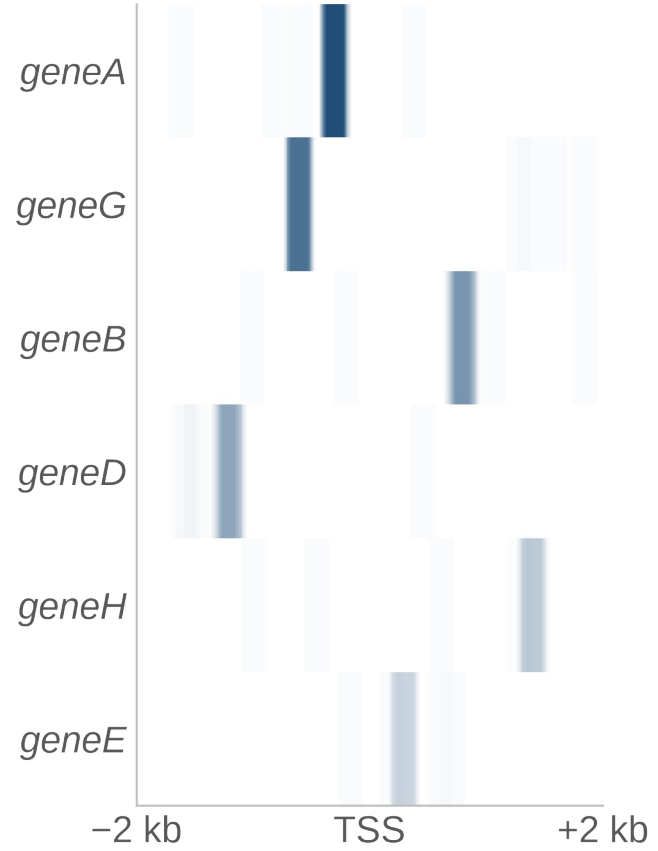


2 average = profile
`plotProfile`



mean of the 6 kept rows
(one sample, toy genes)

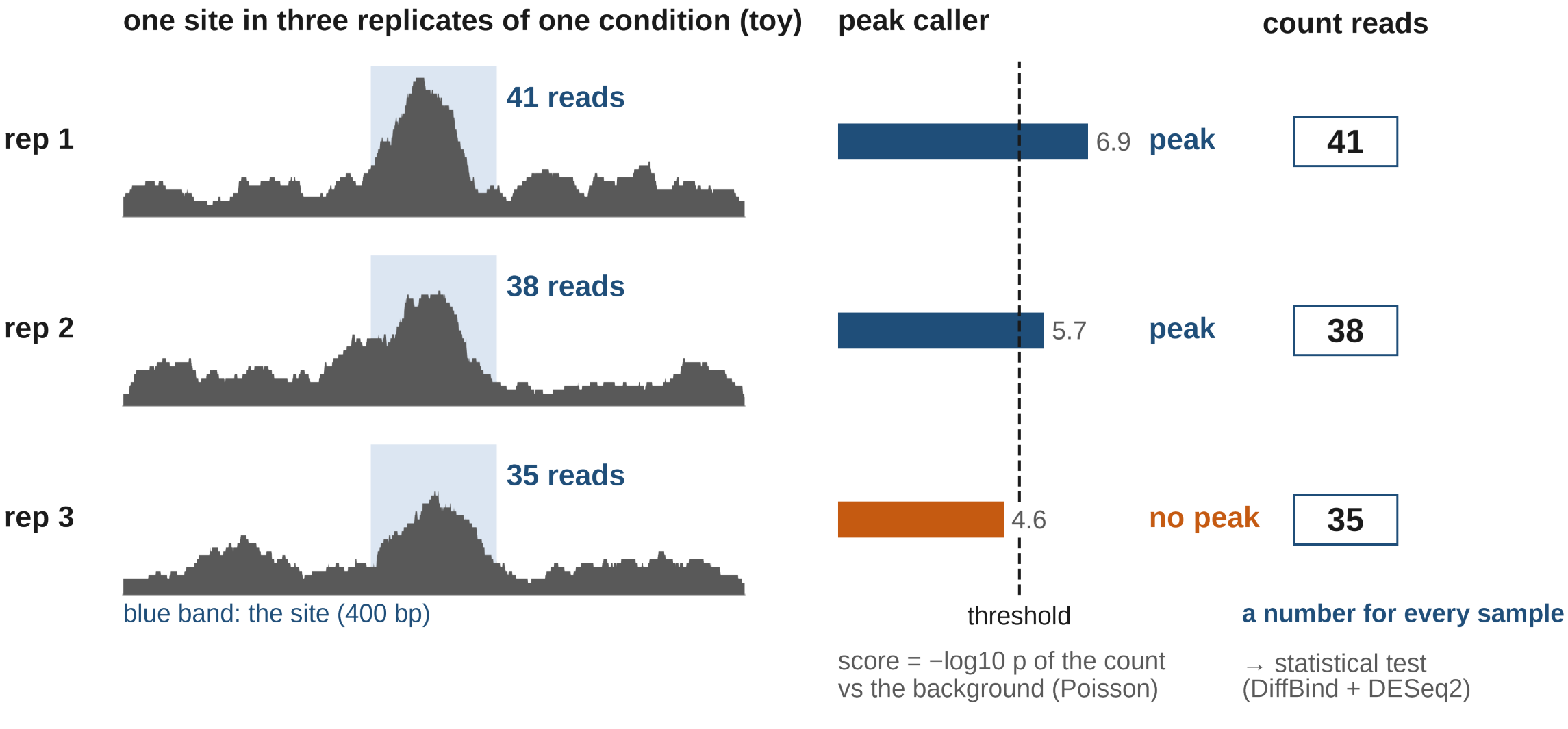
3 rows as colour
`plotHeatmap`



sorted: strongest on top

Part 2 – Differential binding

Why not just count peaks?



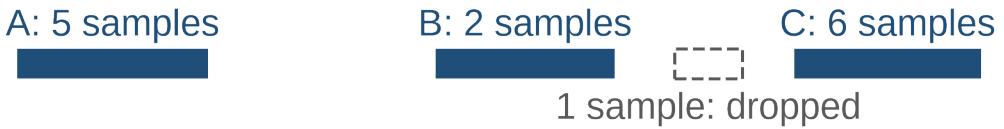
DiffBind: from peaks to a test

1 peaks of each sample



bar = peak
white line = summit

2 consensus



merged; kept if in
≥ 2 samples (minOverlap)

3 summit ± 200 bp



summits = 200:
all sites the same width

4 count the reads of every site in every sample

site	ABA 1	ABA 2	ABA 3	mock 1	mock 2	mock 3
A	88	114	96	36	41	30
B	35	35	30	37	33	47
C	29	43	40	83	75	120

→ normalise by library size → DESeq2 →

log2 FC	FDR
+1.48	2.4e-06
-0.22	1.00
-1.31	2.2e-04

one test per site

The sample sheet

```
SampleID,Condition,Replicate,bamReads,ControlID,bamControl,Peaks,PeakCaller
ABA_rep1,ABA,1,bam/ABF3_ABA_rep1.bam,control,bam/control.bam,peaks/ABF3_ABA_rep1_peaks.narrowPeak,narrow
…  ABA_rep2, ABA_rep3
mock_rep1,mock,1,bam/ABF3_mock_rep1.bam,control,bam/control.bam,peaks/ABF3_mock_rep1_peaks.narrowPeak,narrow
```

PeakCaller narrow:
MACS narrowPeak format

SampleID, Condition, Replicate

ABA	1	2	3
mock	**1**	2	3

the groups to compare (ABA vs mock) and the replicates in each

bamReads

`bam/ABF3_ABA_rep1.bam`

…

`bam/ABF3_mock_rep1.bam`

…

mapped reads of this ChIP sample (minimap2 + samtools sort)

ControlID, bamControl

`bam/control.bam`

= control_rep1 + rep2 (samtools merge)

the same control in all six lines

Peaks

`peaks/ABF3_mock_rep1_peaks.narrowPeak`

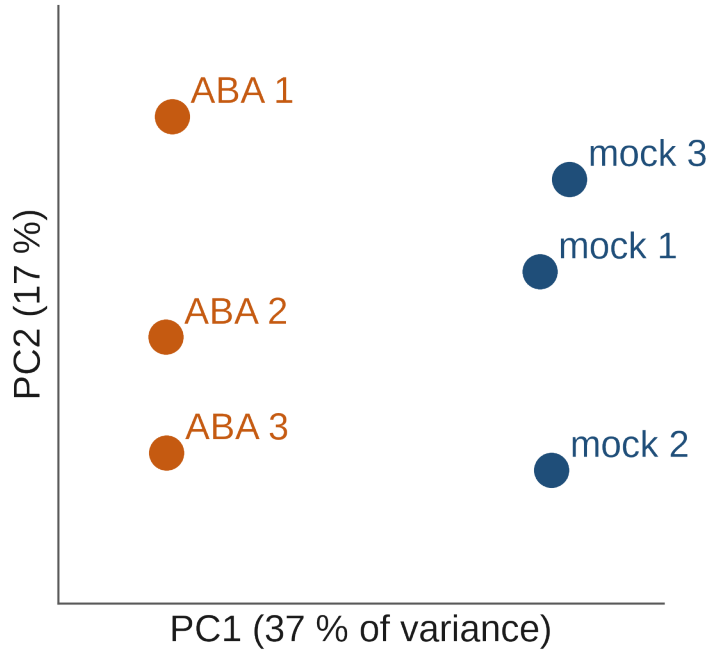


MACS3 peaks of this sample: DiffBind builds the consensus sites from them

6 lines = 6 ChIP samples (3 ABA + 3 mock); the control is not a line of its own

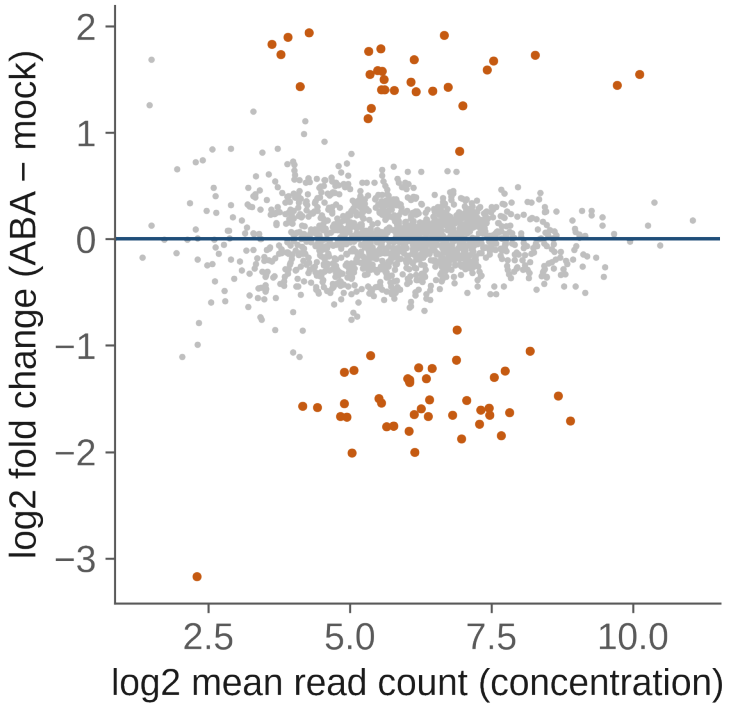
Reading DiffBind plots

PCA (dba.plotPCA)



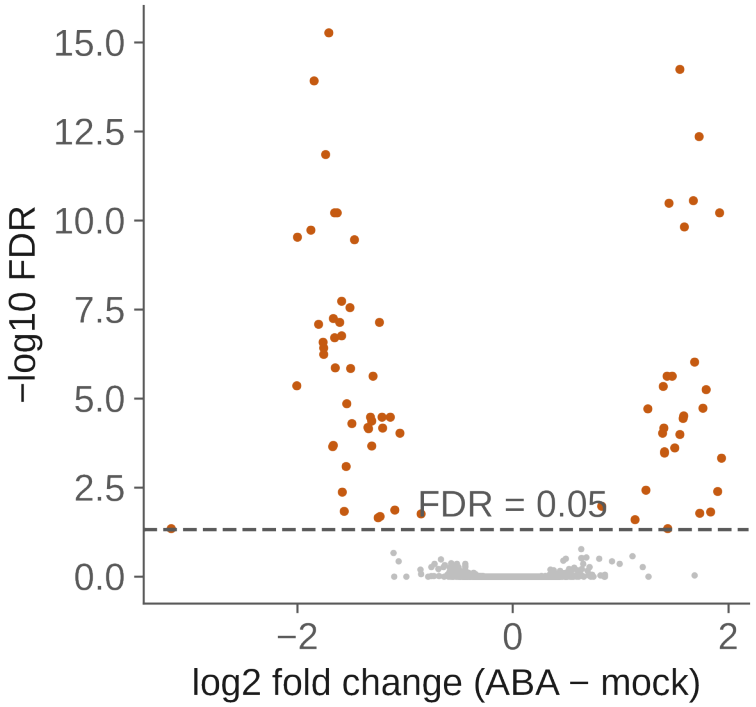
one point = one sample
close = similar read counts over all sites

MA plot (dba.plotMA)



one point = one site
orange: FDR < 0.05
above 0: more reads in ABA

volcano (dba.plotVolcano)

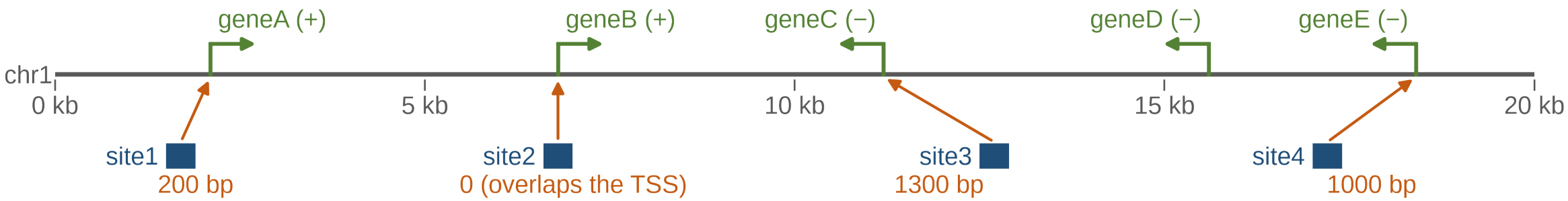


one point = one site
right: stronger in ABA
higher: more significant

bedtools closest: the nearest gene

```
bedtools closest -a sites.bed -b tss.bed -d -t first
```

-d add the distance in bp as the last column -t first if two TSS are equally close, report only the first



output (one line per site):

```
chr1  1500   1901   site1  1.85   0.0004   chr1  2100   2101   TG001  geneA  +  200
chr1  6600   7001   site2  0.42   0.31     chr1  6800   6801   TG002  geneB  +  0
chr1  12500  12901  site3  2.1    0.0001   chr1  11200  11201  TG003  geneC  -  1300
chr1  17000  17401  site4  -0.9   0.04     chr1  18400  18401  TG005  geneE  -  1000
```

A: the site (columns 1–6) **B: the closest TSS (columns 7–12)** **distance (bp)**

Motifs: what is common to the sites?

12 target sequences (sites)

1 CAGTAATA**TTGACT**AGGAAATGATAATTTA
2 TT**AGTCAA**CGATATCTTAATTTACATTATA (− strand)
3 TAGTTGCGTGCCAATAATCTGCTGCTAATG
4 ACAGGATATT**TTGACT**TGAACGTTAATTCT
5 TTGCTGTC**TTGACC**GAACATAGGGAAGGTC
6 AGTCTGCTGAAAAAATAAGTACCAAGTCTT
7 CATTTTTTAGT**AGTCAAGTGAA**AAGAACTA (− strand)
8 AATATACTAAATGATCCT**ATGACT**GGCAAA
9 TCTCGTATA**TTGACT**ATCTATCTCACATTT
10 CTATAAGTAATGCTAACGCATTCGTCCAAA
11 AGACAGTT**AGTCTA**TAGTAG**GGTCAA**GGGT (− strand)
12 ATAAAAATGTAT**TTGACT**TAAATTATTAGA

+ 60 background sequences

(random regions of the same size)

count each 6-mer (both strands)

6-mer	targets		background		p
TTGACT	6/12	50 %	1/60	2 %	4e-5
TGACTA	4/12	33 %	0/60	0 %	5e-4
CTTGAC	3/12	25 %	1/60	2 %	1e-2
AATGAT	2/12	17 %	0/60	0 %	3e-2
ACTAAA	2/12	17 %	0/60	0 %	3e-2
AGACAG	2/12	17 %	0/60	0 %	3e-2

… ranked by enrichment p-value

(hypergeometric test)

words ≤ 1 mismatch from TTGACT

TTGACT
TTGACT
TTGACT
TTGACC
TTGACT
TTCACT
ATGACT
TTGACT
TAGACT
TTGACC
TTGACT

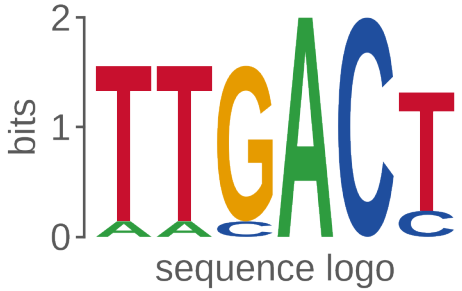
→

	1	2	3	4	5	6
A	1	1	0	11	0	0
C	0	0	1	0	11	2
G	0	0	10	0	0	0
T	10	10	0	0	0	9

counts (of 11)

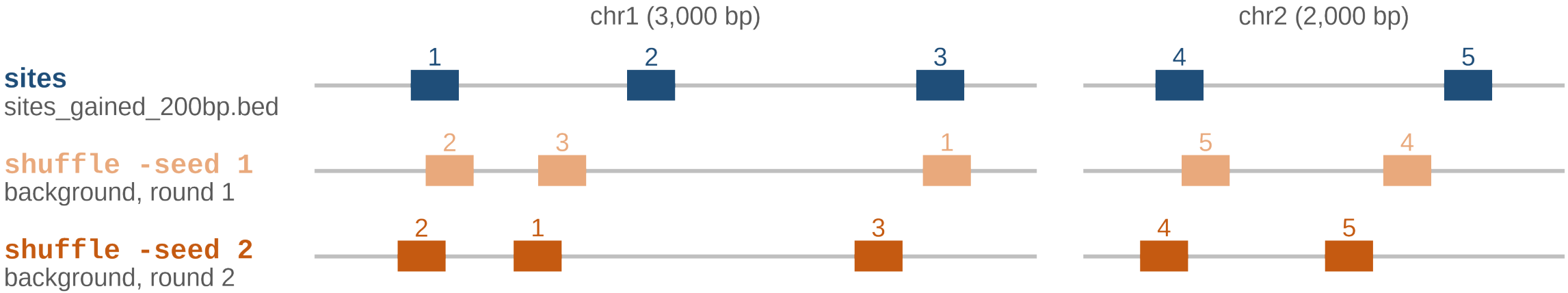
TTGACT: in 50 % of targets, 2 % of background → motif candidate

HOMER then lets in similar words (mismatches) while the enrichment improves → a motif matrix



sequence logo

Background: random regions like the sites



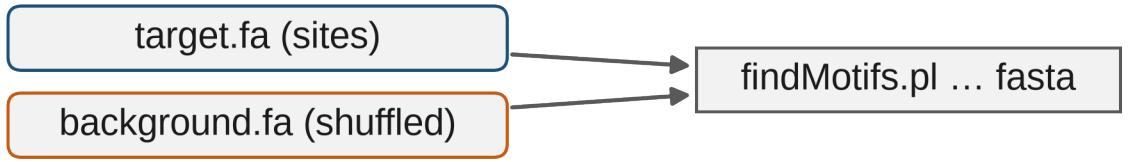
same number, same length (200 bp), same chromosome (-chrom), not overlapping each other (-noOverlapping)

```
bedtools getfasta -fi genome.fa -bed regions.bed
```

```
chr1   400   600   site1   1.0   .
```
one BED line (0-based start, end not included)

→

>chr1:400-600
ATTTTCATAAGCATAGTTAAACCTTTTTAGTAAGTGAGGATAGTTTTCGCTAAT…
one FASTA record: name = chrom:start-end, 200 bp on one line



in the exercise: seeds 1 … 5 → 5 × as many background regions

Reading HOMER results

homerResults.html Total target sequences = 12 Total background sequences = 56

Rank	Motif	P-value	log P-pvalue	% of Targets	% of Background	STD (Bg STD)	Best Match/Details
1 *		1e-5	-1.296e+01	**66.67%**	**4.91%**	4.7bp (5.7bp)	ToyTF/made-up-example/Homer(0.987) **similarity score 0.987** More Information
2 *		1e-2	-5.960e+00	**33.33%**	**3.21%**	9.6bp (3.5bp)	ToyTF/made-up-example/Homer(0.345) **similarity score 0.345** More Information

sorted by p-value

the motif found de novo (logo)

enrichment in targets vs background

natural log of p

targets with the motif

background sequences with motif

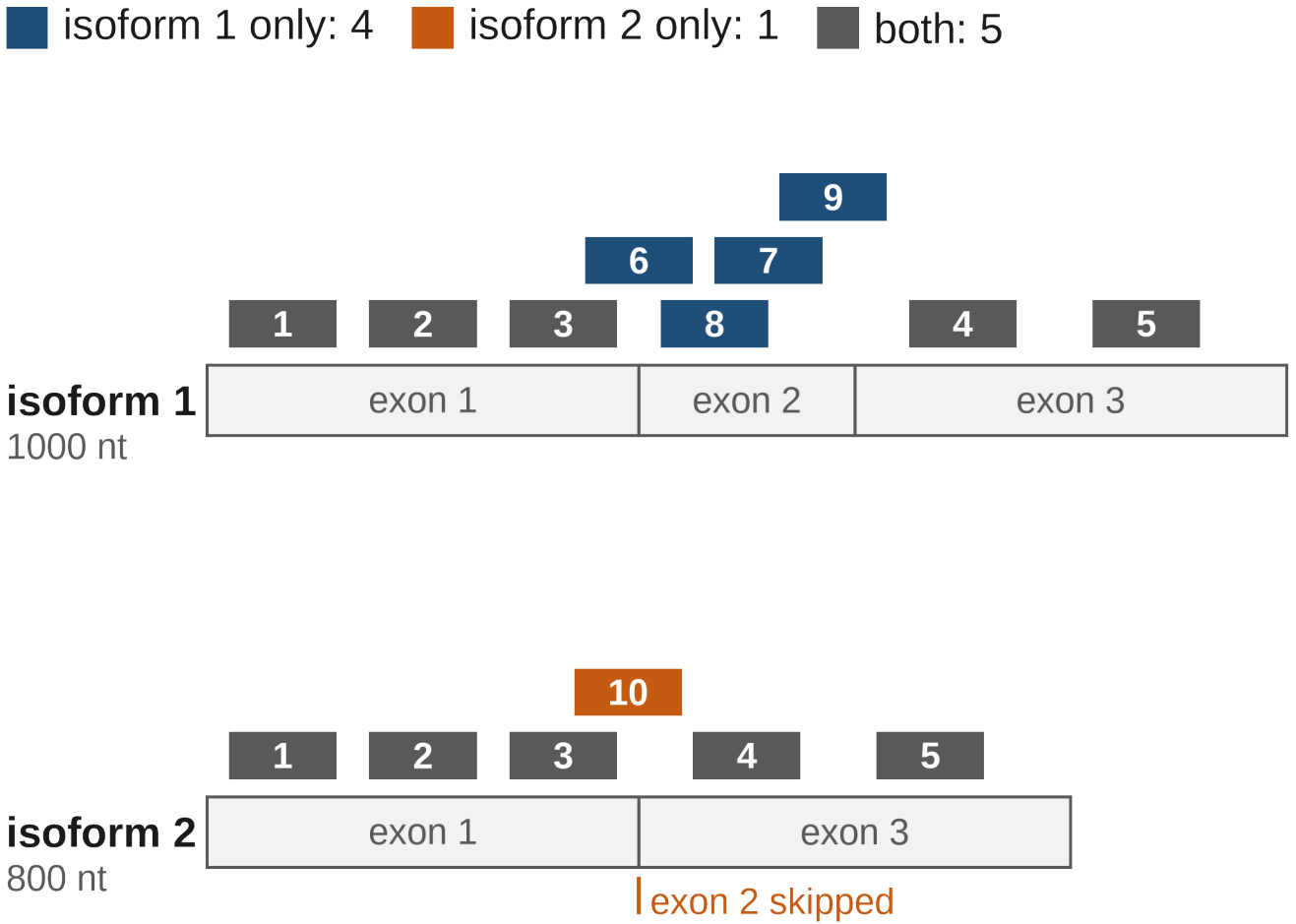
most similar KNOWN motif (here a made-up library with one motif, ToyTF)

score 0–1, 1 = identical: a suggestion, not an identification — compare the logos (More Information)

* = HOMER's flag "possible false positive"

Part 3 – Binding and expression

salmon: reads to transcripts



estimated reads (EM)

round	isoform 1	isoform 2
start	6.50	3.50
1	6.99	3.01
2	7.25	2.75
3	7.39	2.61
…		
35 (stable)	7.57	2.43

start: shared reads split half and half
each round: split by reads per nt
of each isoform (isoform 1 is longer)

gene = 7.57 + 2.43 = 10 reads

Patro et al. 2017, Nat Methods 14:417

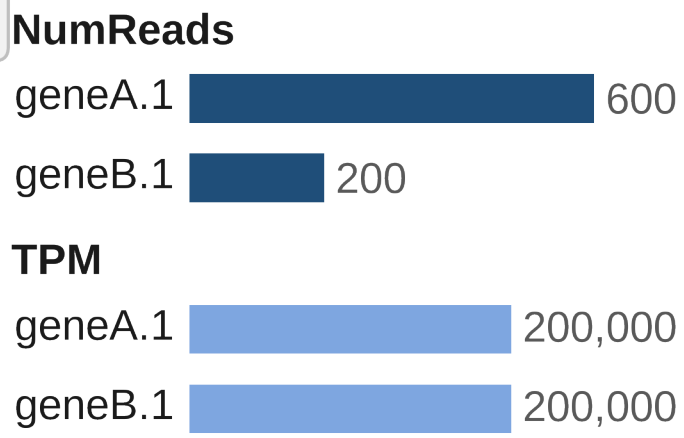
quant.sf and TPM

transcript | length (nt) | corrected for fragment length | transcripts per million | estimated reads

Name	Length	EffectiveLength	TPM	NumReads
geneA.1	3250	3000.000	200000.000000	600.000
geneB.1	1250	1000.000	200000.000000	200.000
geneC.1	750	500.000	600000.000000	300.000

quant.sf

Name	NumReads ÷ EffectiveLength	rate	rate ÷ S × 10^6 = TPM
geneA.1	600 ÷ 3000	0.2	200,000
geneB.1	200 ÷ 1000	0.2	200,000
geneC.1	300 ÷ 500	0.6	600,000
		S = 1.0	**sum = 1,000,000**



geneA.1: 3× the effective length
3× more reads, same TPM

DESeq2: which genes changed?

counts per gene

	ABA		mock	
gene	**1**	**2**	**1**	**2**
geneA	652	1,187	205	138
geneB	180	2,300	330	60
geneC	790	1,520	1,010	690
geneD	41	77	48	36
geneE	2,410	4,480	3,050	2,080
all reads	461k	871k	587k	394k
size factor	**0.84**	**1.56**	**1.06**	**0.73**

median of ratios to the average sample, over all genes (toy: 505 genes)

÷ →

normalised counts

	ABA		mock	
	1	**2**	**1**	**2**
geneA	778	763	194	189
geneB	215	1,478	313	82
geneC	943	977	956	945
geneD	49	49	45	49
geneE	2,877	2,880	2,888	2,848

geneA: replicates agree

geneB: replicates disagree

same fold change, different padj

→

DESeq2 output, ABA vs mock

	baseMean	**log2FoldChange**	**pvalue**	**padj**
geneA	481	2.01	2e-18	8e-16
geneB	522	2.10	0.14	0.99
geneC	955	0.01	0.94	0.99
geneD	48	0.06	0.90	0.99
geneE	2,873	0.01	0.98	0.99

baseMean: mean of the normalised counts

log2FoldChange ≈ log2(ABA ÷ mock)

padj: pvalue corrected for testing all genes (FDR)

Love et al. 2014, Genome Biol 15:550

Binding and expression together

binding: sites + closest TSS (toy)

site	gene	distance	log2FC	FDR	group
s1	geneA	120	1.8	0.001	gained
s2	geneB	450	2.1	0.010	gained
s3	geneB	800	0.2	0.600	(see s2)
s4	geneC	300	0.1	0.700	unchanged
s5	geneD	2,400	1.5	0.002	too far
s6	geneE	90	-1.2	0.020	lost: none
s7	geneF	600	0.3	0.400	unchanged
s8	geneG	50	1.1	0.030	gained

keep distance ≤ 1,000
gained: FDR < 0.05 and log2FC > 0
unchanged: FDR ≥ 0.05, gene not gained

expression: DESeq2 (toy)

gene	log2FC	padj	class
geneA	2.5	0.001	induced
geneB	0.4	0.300	n.s.
geneG	-1.1	0.010	repressed
geneC	1.2	0.020	induced
geneF	0.1	0.900	n.s.

+ 7 other genes

induced: padj < 0.05, log2FC > 0
repressed: padj < 0.05, log2FC < 0

Is the share of induced genes higher with gained binding?
→ your data, Part 3.3

% of genes (toy)

